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Therapy of psoriasis

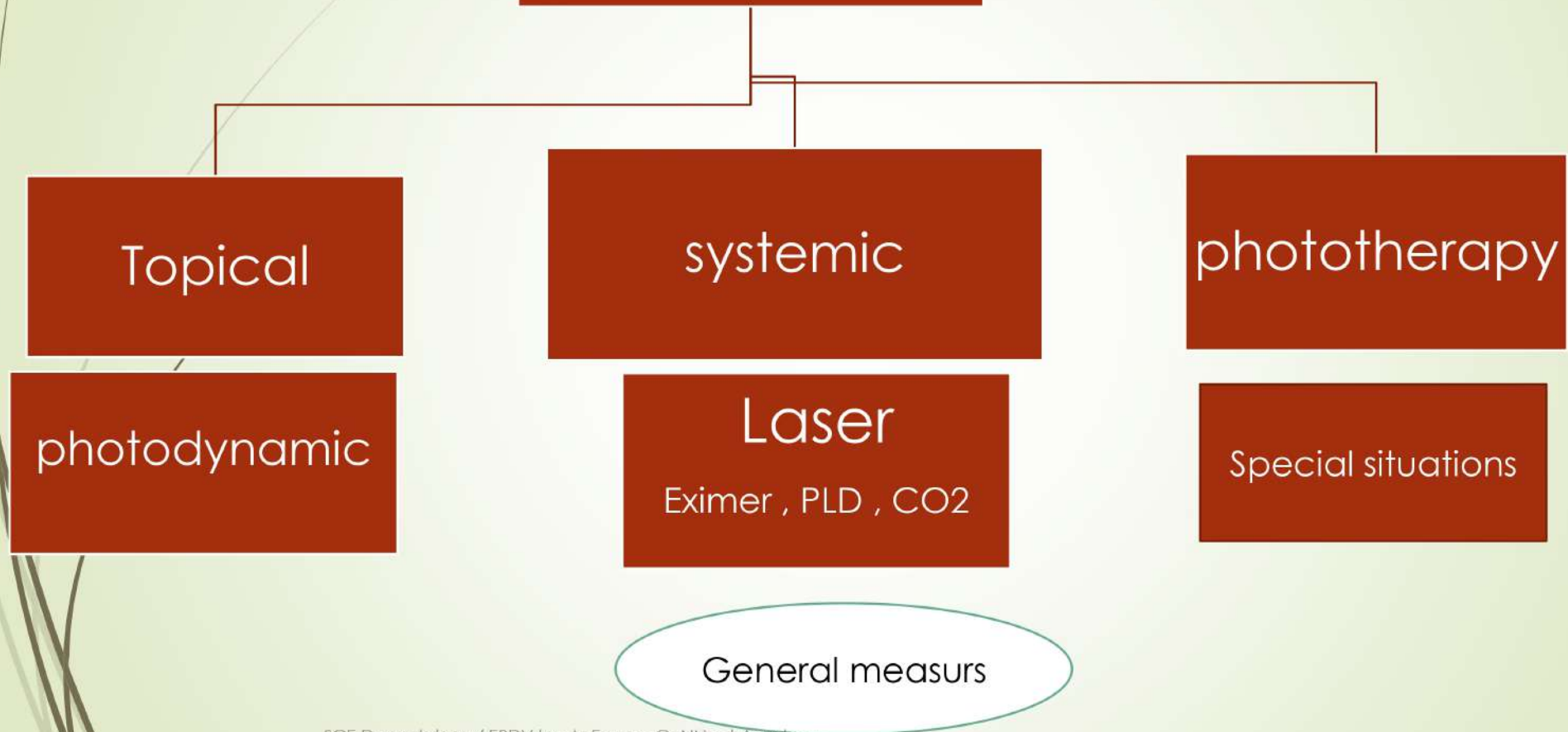
dr eman goma

Consultant dermatology

SCE Dermatology / EBDV by dr:Eman - CoNNeCT Academy



Treatment of ps



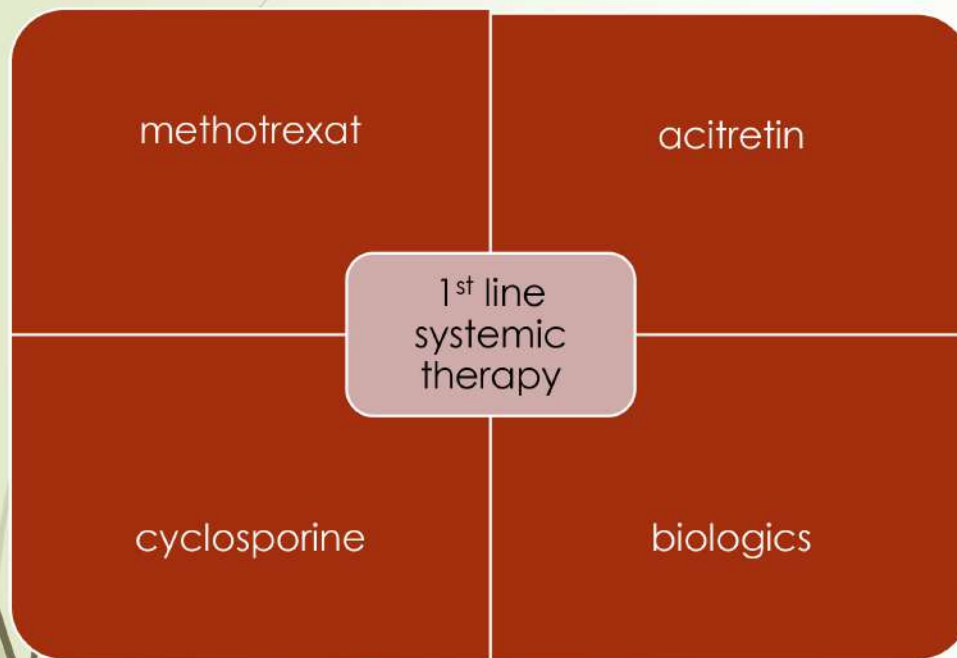
Systemic therapies

- General indications for systemic therapies in psoriasis:

severe forms of psoriasis including

1. Chronic plaque psoriasis of large surface area ($>10-15\%$) or not responding to topical therapies.
2. Erythrodermic psoriasis.
3. Pustular psoriasis.
4. Moderate to severe psoriatic arthropathy.
5. Severe nail psoriasis.
6. Psoriasis of small surface area but impairing quality of life e.g. severe palmoplantar or scalp psoriasis

Systemic therapies available include



Other (second-line) systemic therapies:

5. Azathioprine
6. Mycophenolate mofetil
7. Fumaric acid esters (also known as fumarates)
8. Hydroxyurea
9. Lefhmomide
10. Sulfasalazine
11. Tacrolimus
- 12.6-Thioguanine

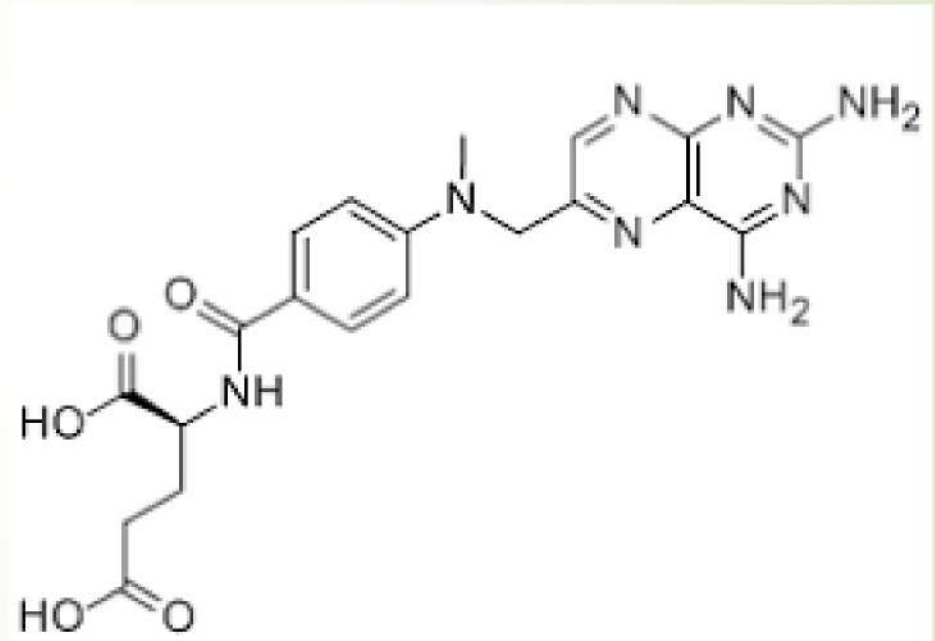
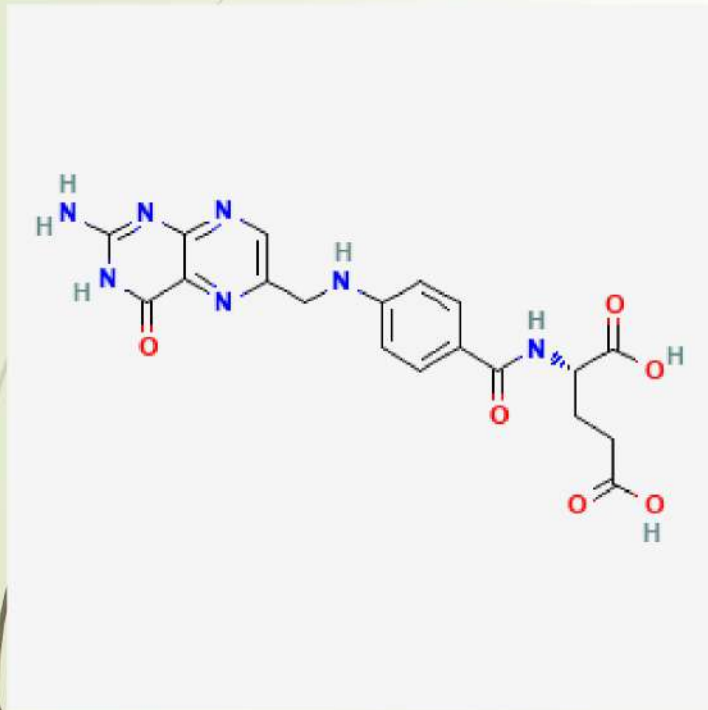
	METHOTREXAT	ACETRITINE	CYCLOSPORINE
DOSE	Single oral/IM weekly dose (7.5 - 25 mg) + oral folic acid 1-5 mg daily except for the day of methotrexate	10-50 mg/day given as a single dose for 3 - 6 months	2.5-5 mg/kg/day in divided doses (for maximum 1 - 2 years)
S . E	Nausea / vomiting, Hepatotoxic, Bone marrow suppression, immuno-suppression, teratogenic and pulmonary fibrosis	Teratogenic Hypelipidemia Cheilitis/Xerosis Hepatotoxic	Nephrotoxic, Hypertension, Malignancy, immunosuppression, Hypertrichosis, Gingival hyperplasia
contraindication	Pregnancy (X) Lactation Liver/kidney disease Anaemia, leucopenia or thrombocytopenia Active infection Hypersensitivity to	Pregnancy (X) Lactation Hypelipidemia	Abnormal renal function Uncontrolled hypertension Malignancy Hypersensitivity to Cyclosporine Live vaccinations o

monitoring	CBC Liver/kidney function tests Pregnancy test Tests for HIV/PPD Chest x-ray Liver biopsy	Pregnancy test Lipid profile CBC Liver/kidney function tests	Blood pressure Kidney function tests (BUN, creatinine and Urine analysis) Lipid profile

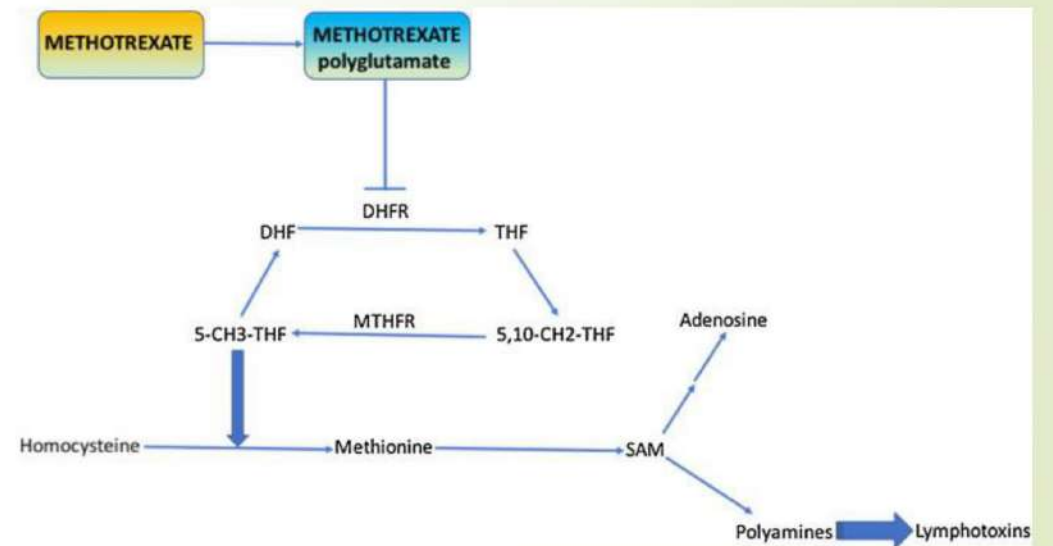
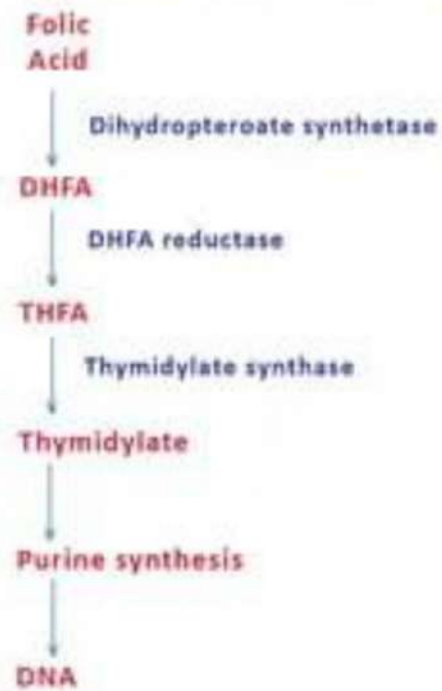
methotrexate



Folic acid versus methotrexat



MECHANISM OF ACTION



INDICATIONS AND CONTRAINDICATIONS FOR METHOTREXATE

Indications

- Severe psoriasis
 - Chronic plaque psoriasis (>10–15% BSA or interference with employment or social functioning)
 - Pustular psoriasis (generalized or localized)
 - Erythrodermic psoriasis
 - Psoriatic arthritis (moderate to severe)
 - Severe nail psoriasis
- Psoriasis not responding to topical treatments, photo(chemo)therapy and/or systemic retinoids

Contraindications (absolute or relative)

- Severe anemia, leukopenia, and/or thrombocytopenia*
- Significant liver function abnormalities, hepatitis (active and/or recent), severe fibrosis, cirrhosis, excessive alcohol intake*
- Impaired kidney function (creatinine clearance <60 ml/min)[†]
- Concomitant medications that increase MTX levels, e.g. trimethoprim-sulfamethoxazole*
- Significantly reduced pulmonary function*
- Pregnancy or lactation*
- Currently planning to have children (male and female patients)[‡]
- Severe infections* or active infections
- Peptic ulcer (active)* or gastritis
- Hypersensitivity to MTX*
- Unreliable patient*
- Pleural effusion or ascites[†]
- Concomitant hepatotoxic medications
- Immunodeficiency syndromes
- Concomitant radiation therapy

*Absolute contraindication. CE Dermatology / EBDV by dr:Eman - CoNNect Academy

[†]Requires significant reduction in dosage.

[‡]Because of possible mutagenic risk and teratogenicity, discontinue MTX 3 months prior to attempts to conceive; continue contraception during these 3 months

Subjective symptoms	Hematopoietic [†]	Mucocutaneous
<i>Most frequent</i> *, [†] Nausea Vomiting Abdominal pain Fatigue Headache <i>Occasional</i> Loss of libido Impaired memory	Leukopenia [†] Thrombocytopenia Anemia	Oral erosions [†] Alopecia Delayed phototoxicity Tenderness and/or necrosis of plaques due to overdose Rarely, urticaria, angioedema, vasculitis
Hepatic	Pulmonary	Fetal development
Hepatitis [†] Cirrhosis [§]	Interstitial pneumonitis (acute-onset cough, dyspnea)	Birth defects: cranial and absence of digits
Carcinogenesis [¶]	Idiosyncratic reaction	
In rheumatoid arthritis patients, increased risk of lymphoma Risk factor for cutaneous SCC in PUVA-treated patients	Early in course with full dosage Severe pneumonia Gastrointestinal hemorrhage Pancytopenia	
Opportunistic infections	Additional/unusual	
<i>Pneumocystis jiroveci</i> pneumonia Cryptococcosis Disseminated	Osteopathy (pain, osteoporosis, compression fractures) Ventricular cardiac arrhythmias Lowered seizure threshold	

acitretine



INDICATIONS AND CONTRAINDICATIONS FOR ACITRETIN

Indications

- Severe psoriasis that cannot be managed by topical treatments or photo(chemo)therapy
- Monotherapy is indicated for erythrodermic or pustular psoriasis
- Combination therapy is indicated for chronic plaque psoriasis

Contraindications

- Severe liver dysfunction*
- Severe kidney dysfunction (elimination reduced)*
- Pregnancy or lactation*
- Women of childbearing potential who cannot guarantee adequate contraception during and up to 3 years following discontinuation of acitretin*
- Hyperlipidemia, especially hypertriglyceridemia, that cannot be controlled*
- Excessive alcohol intake*
- Unreliable patient*
- Concomitant medications that interfere with retinoid bioavailability or whose metabolism is altered†
- Concomitant hepatotoxic drugs, e.g. methotrexate‡
- Poorly controlled diabetes mellitus
- History of pancreatitis
- Use of contact lenses
- Atherosclerosis

*Absolute contraindications.

†Requires dose adjustment and careful monitoring, e.g. phenytoin competes for plasma protein binding.

‡Reserved for treatment-resistant patients.

cyclosporine



INDICATIONS AND CONTRAINDICATIONS FOR CYCLOSPORINE

Indications

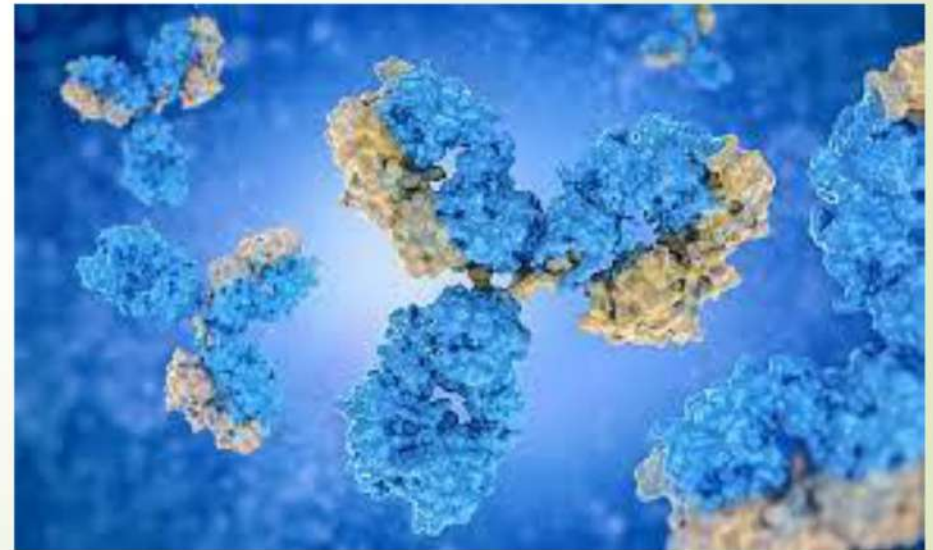
- Severe psoriasis
- Conventional therapies (topical treatments, photo(chemo)therapy, acitretin, methotrexate) ineffective or inappropriate

Contraindications

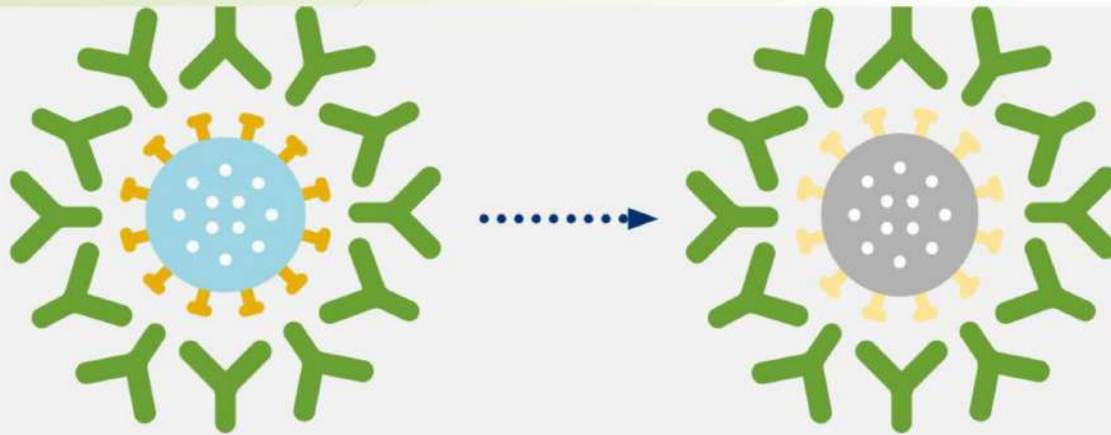
- Impaired renal function*
- Uncontrolled hypertension*
- Past or present* malignancy
- History of excessive photo(chemo)therapy (>200 PUVA treatments) or concurrent photo(chemo)therapy*
- Concomitant drugs affecting cyclosporine pharmacokinetics†
- Radiotherapy*
- Severe infections* or active infections
- Hypersensitivity to cyclosporine*
- Unreliable patient*
- Primary or secondary immunodeficiency
- Concomitant immunosuppressive therapy
- History of arsenic exposure
- Pregnancy or lactation
- Concurrent methotrexate administration
- Significant hepatic disease
- Hyperuricemia, hyperkalemia
- Vaccination with live vaccines
- Seizure disorder
- Poorly controlled diabetes mellitus
- Severe chronic organ dysfunction
- Alcohol and drug abuse
- Malabsorption

4- biologics targeted therapy

- It is protein molecules derived from living organism to target specific protein in immune system
- Types
 - Monoclonal antibody (mab)
 - Fusion antibody protein (cept)
 - Recombinant human cytokins

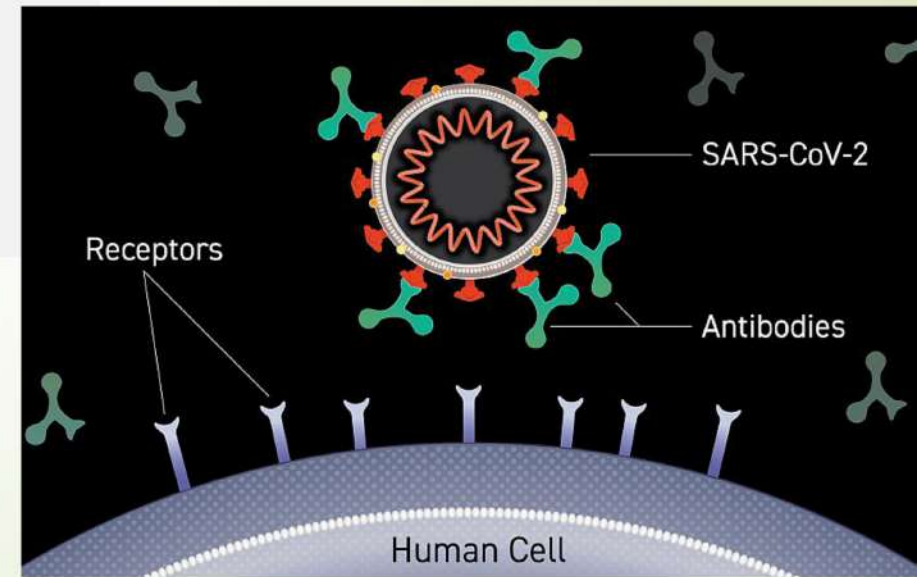


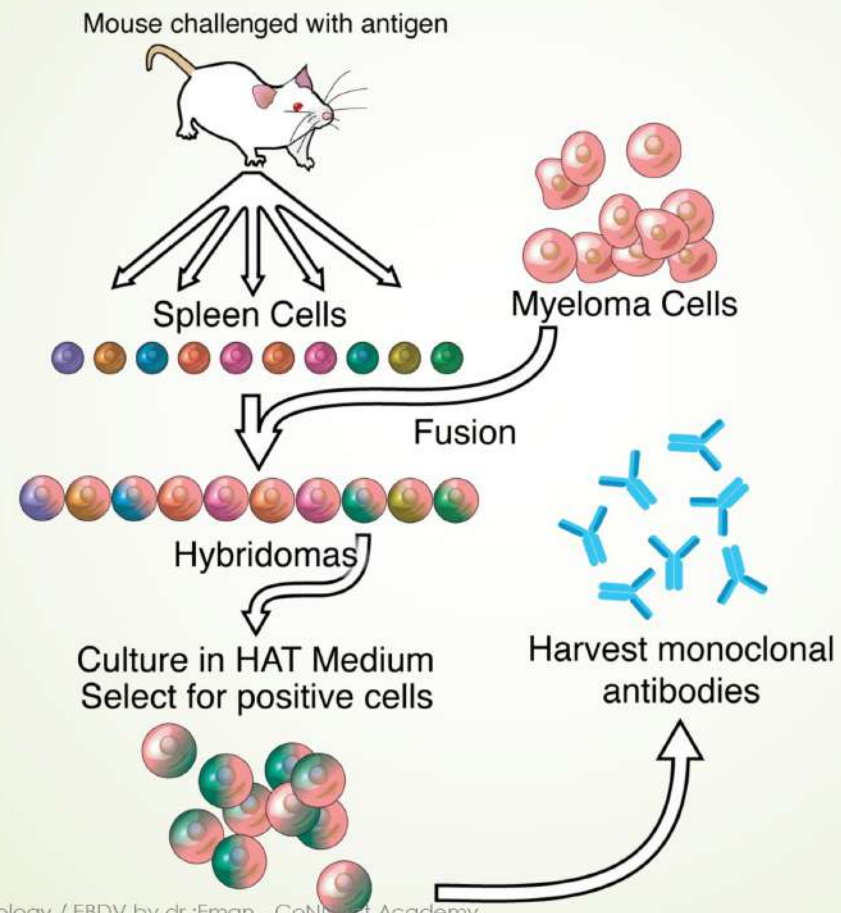
Monoclonal antibody



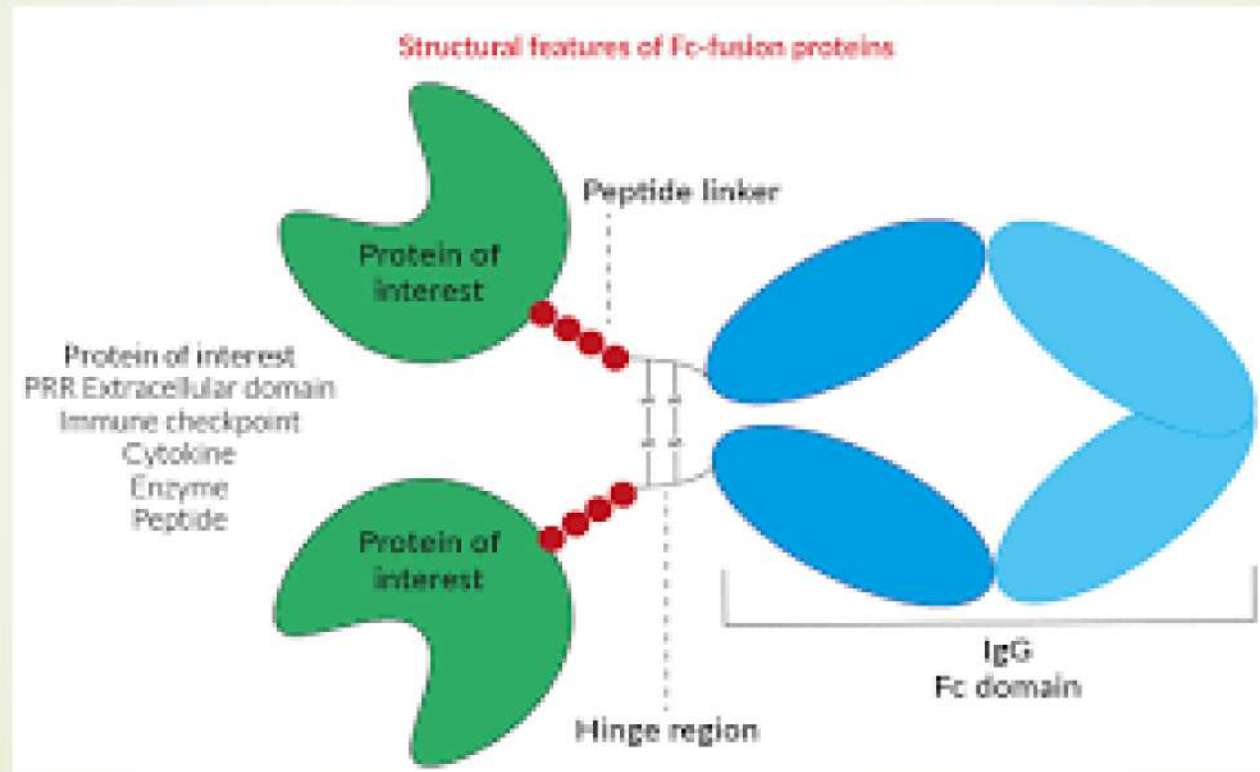
**Monoclonal antibodies
bind to their target**

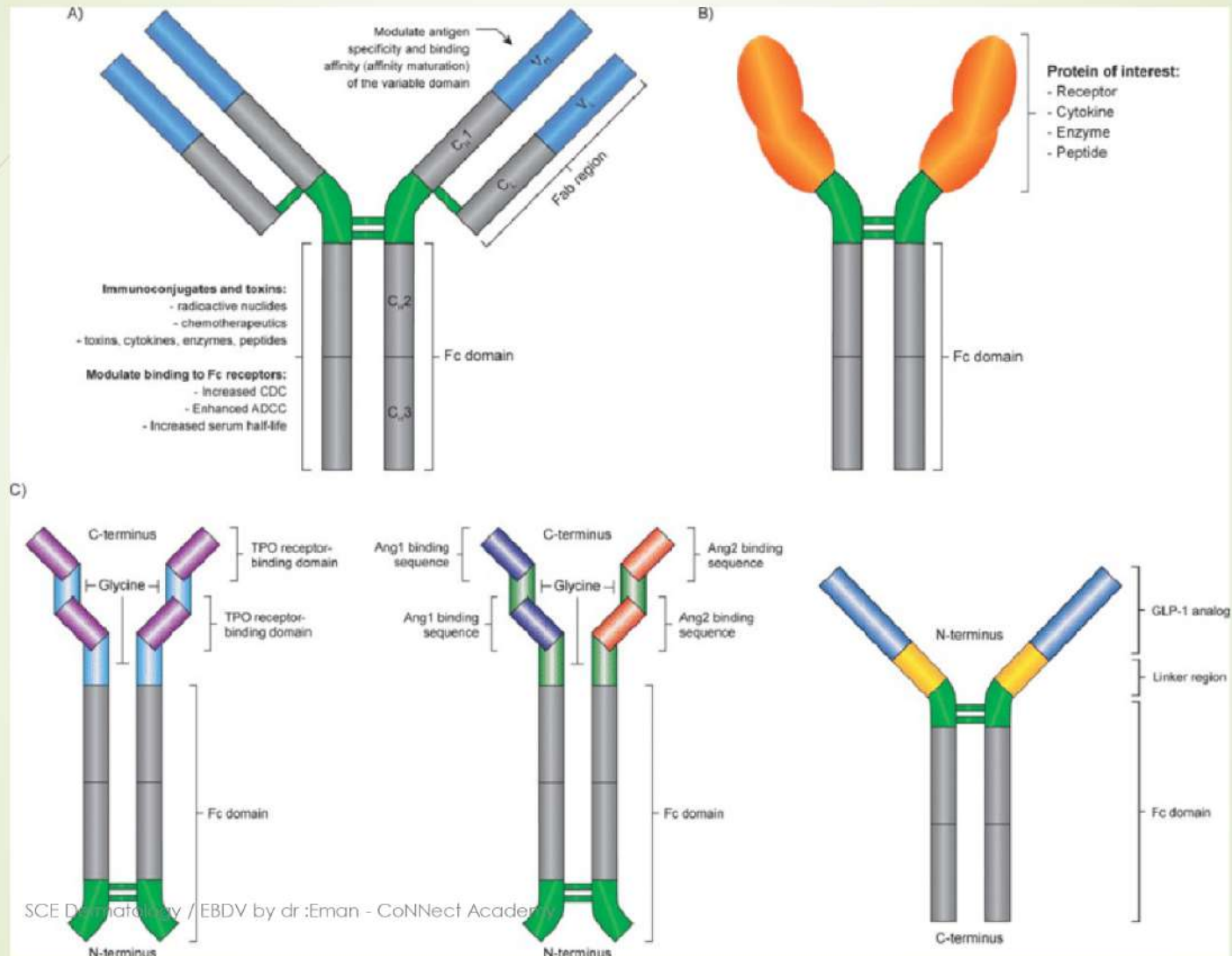
**once attached,
they make it harmless**





Fusion antibody protein





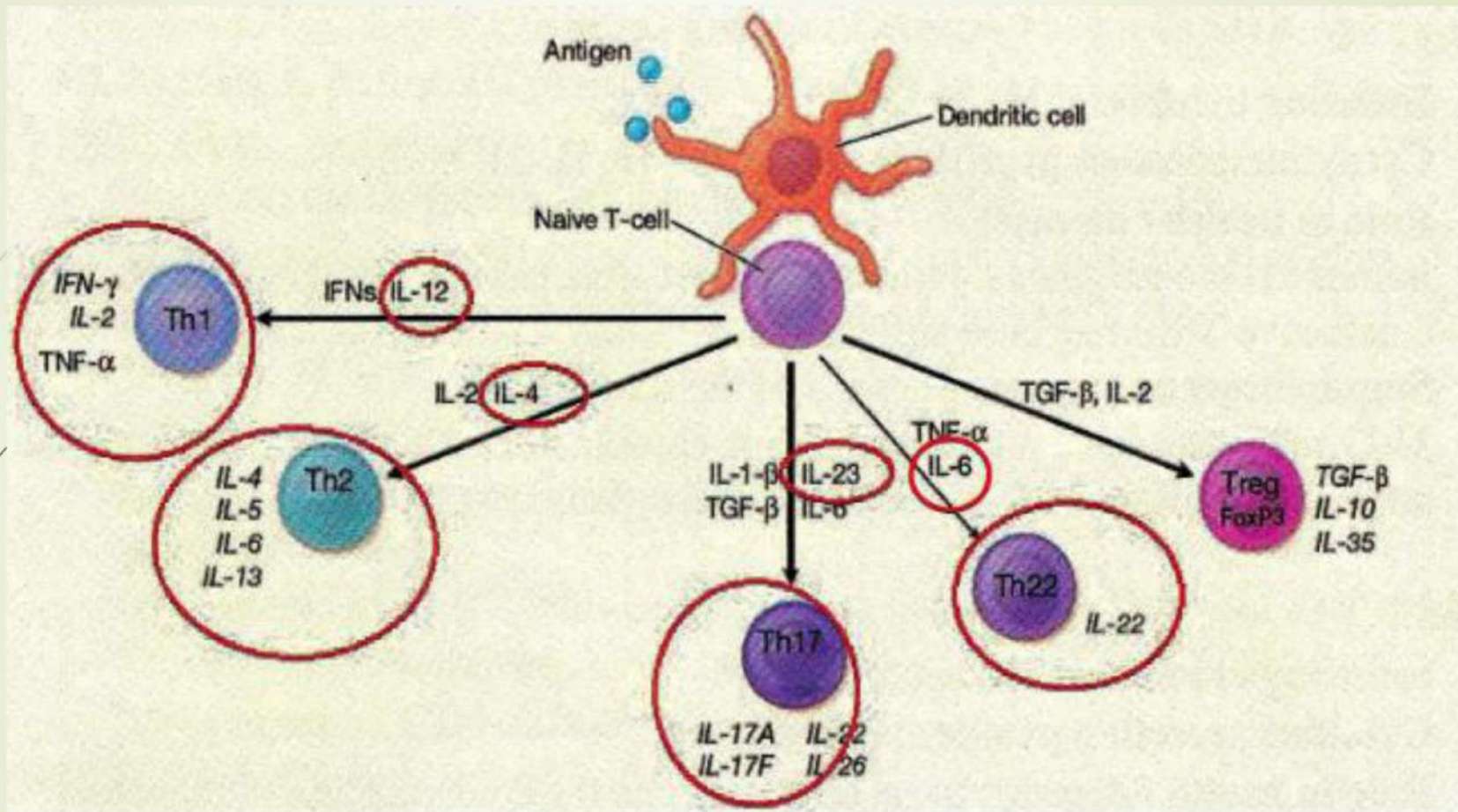
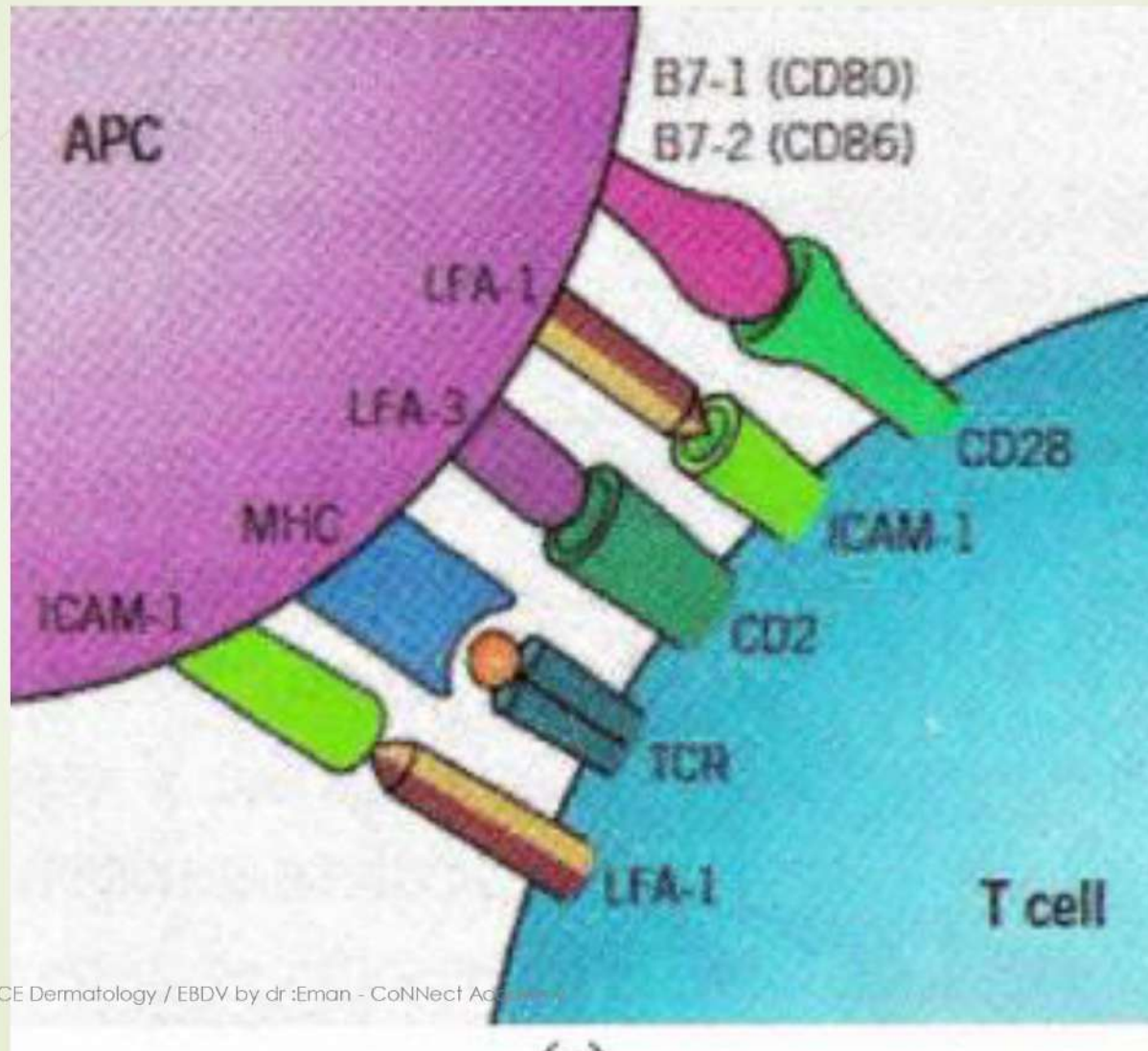


Fig 3.5 Differentiation of naïve Th cells (modified from *Fitzpatrick, 2012*).



COMMERCIALLY AVAILABLE BIOLOGIC AGENTS FOR THE TREATMENT OF PSORIASIS

Biologic agent	Target	Molecule	Approved*
Etanercept	TNF- α [†]	Human fusion protein	FDA + EMA
Infliximab	TNF- α [‡]	Chimeric antibody	FDA + EMA
Adalimumab	TNF- α [‡]	Human antibody	FDA + EMA
Ustekinumab	p40 subunit of IL-12/23	Human antibody	FDA + EMA
Secukinumab	IL-17A	Human antibody	FDA + EMA
Ixekizumab	IL-17A	Humanized antibody	FDA + EMA
Brodalumab	IL-17 receptor	Human antibody	FDA + EMA
Guselkumab	IL-23	Human antibody	FDA

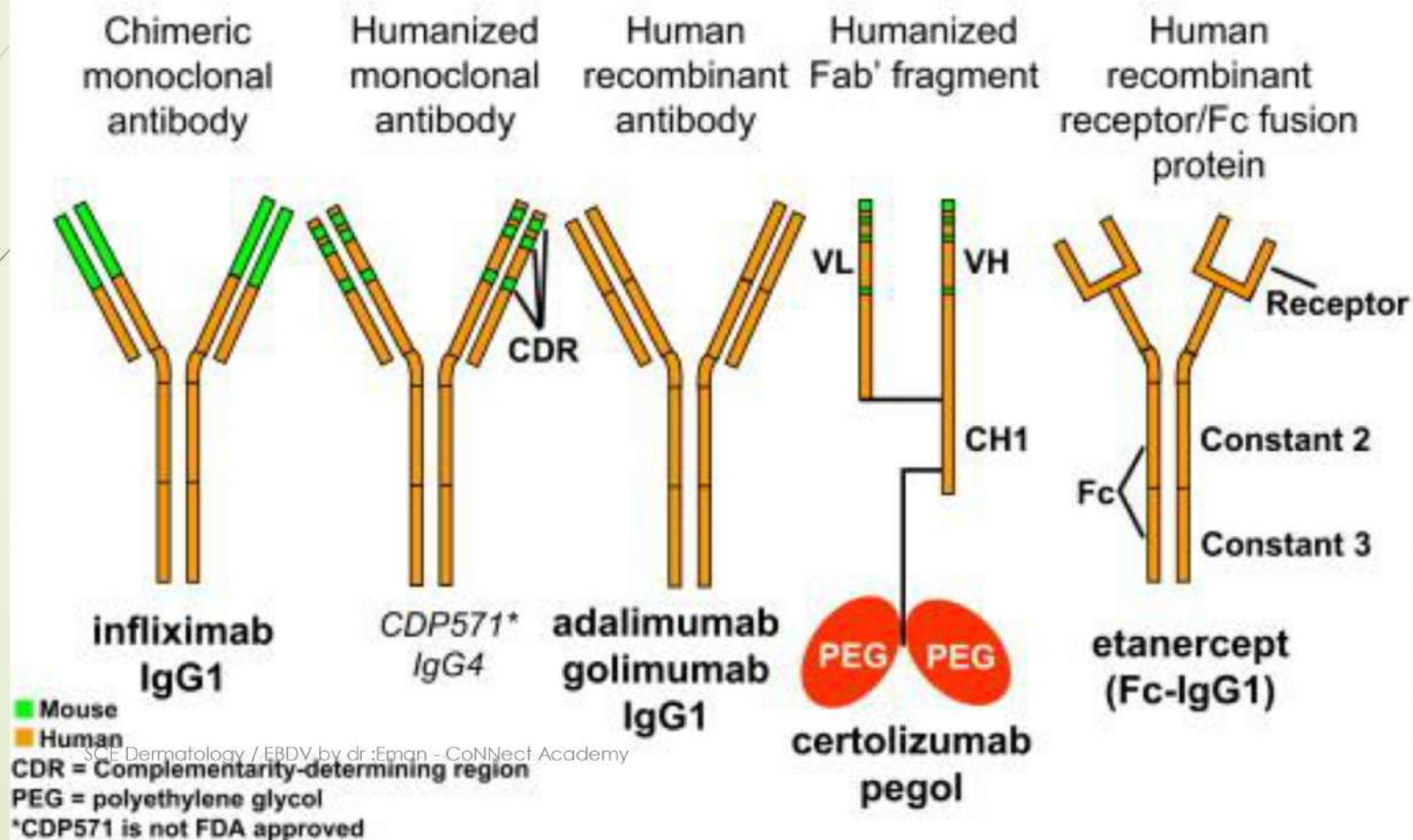
*By registration authorities for the treatment of psoriasis as of July 2017. Biosimilar products, e.g. adalimumab-atto, etanercept-szys, infliximab-dyyb, are also commercially available.

[†]Soluble form.

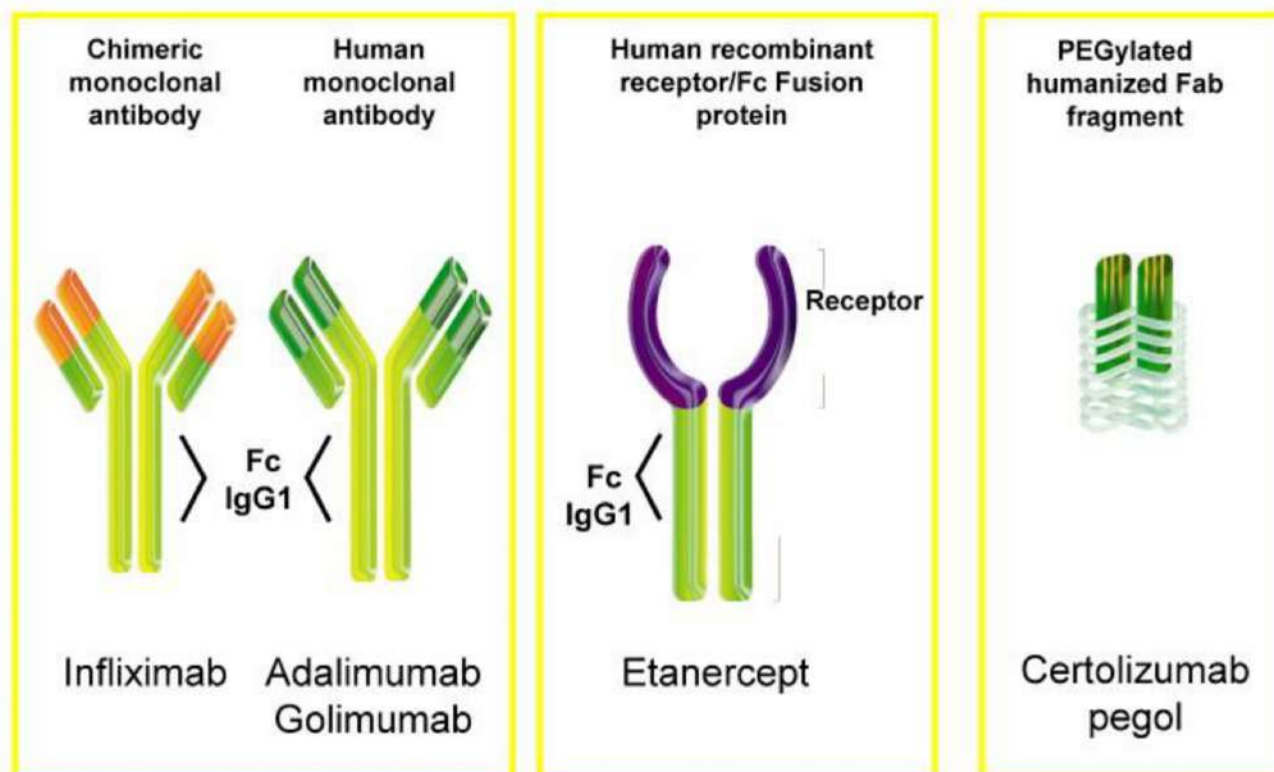
[‡]Soluble form and transmembrane TNF receptor.

Target	Biologics	Notes
CD2 (on T cells)	Alefacept	Prevent interaction with <i>LFA-3</i> (on APCs)
CD11a (a component of LFA-1 on T cells)	Efalizumab (withdrawn from the market in 2009)	Prevent interaction with <i>ICAM-1</i> (on APCs)
TNF-α MCQ	<i>TNF-α inhibitors</i> (etanercept, infliximab, adalimumab)	
IL-12 and IL-23	Ustekinumab	IL-12 → development of Th1 IL-23 → development of Th17
IL-17 A	Secukinumab, ixekizumab	
IL-17RA “receptor”	Brodalumab	

TNF Inhibitors



Monoclonal antibodies, fusion protein and PEGylated Fab fragment directed against TNF α



➤ Side effects of biologics:



1. Increased risk of infections or reactivation of latent infections e.g. tuberculosis, bacterial sepsis, systemic fungal infections (e.g. histoplasmosis), hepatitis B virus

2. Could potentially increase the risk of malignancies (especially lymphomas).



3. Hypersensitivity reactions (e.g. urticaria, angioedema up to anaphylaxis).



4. Cardiovascular and CNS complications

➤ **Baseline monitoring and follow-up:**



1. Screening for tuberculosis (PPD - Interferon- γ release assay - chest X-ray)



2. Hepatitis B and C virus serologic profiles



3. HIV testing



4. CBC



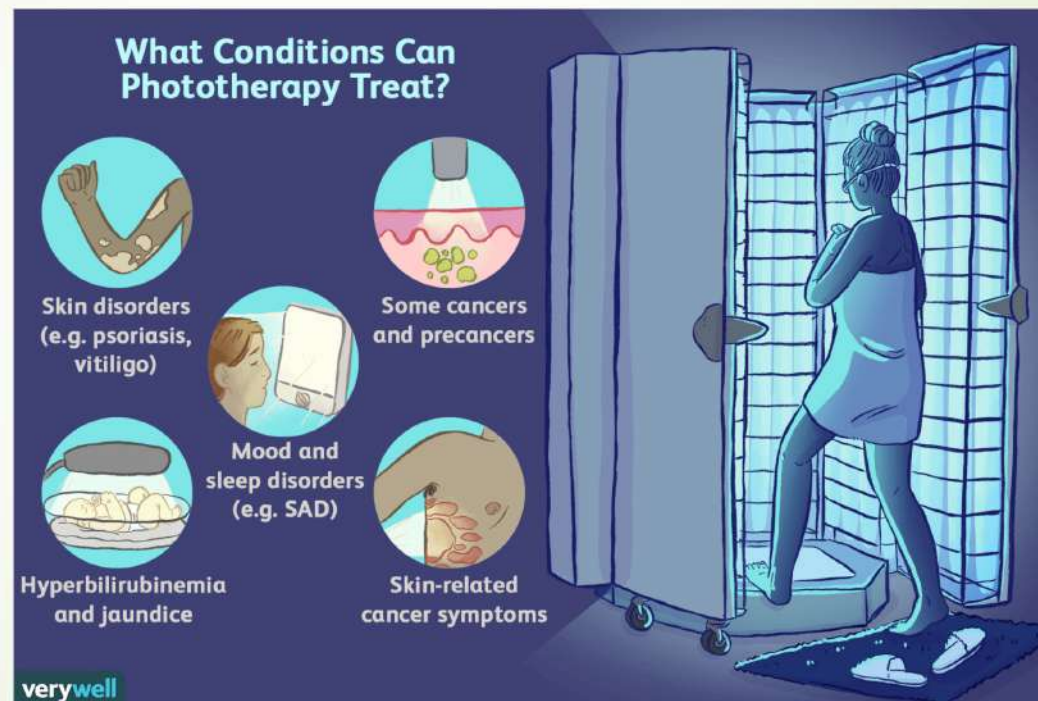
5. Kidney / liver function tests

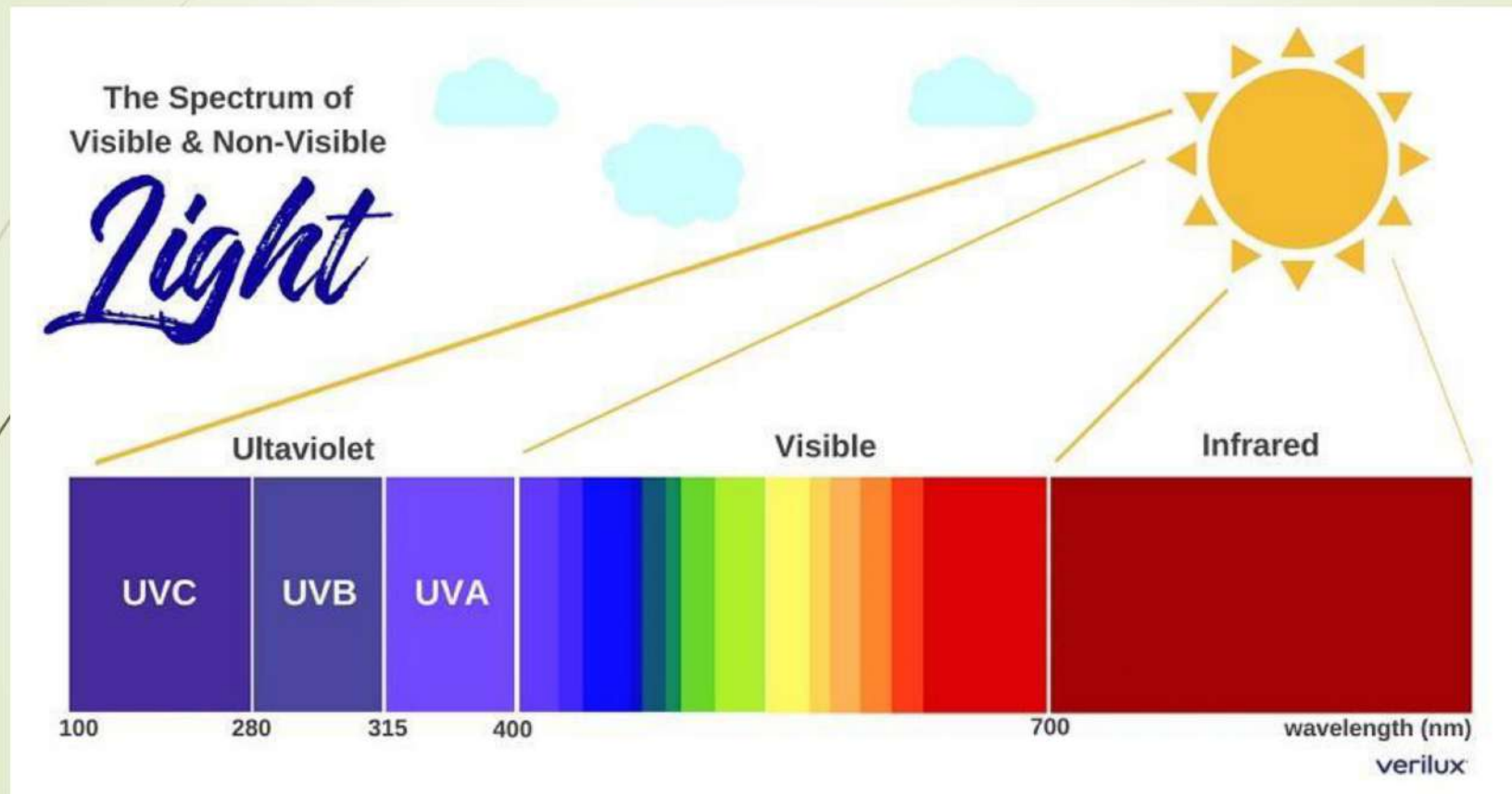
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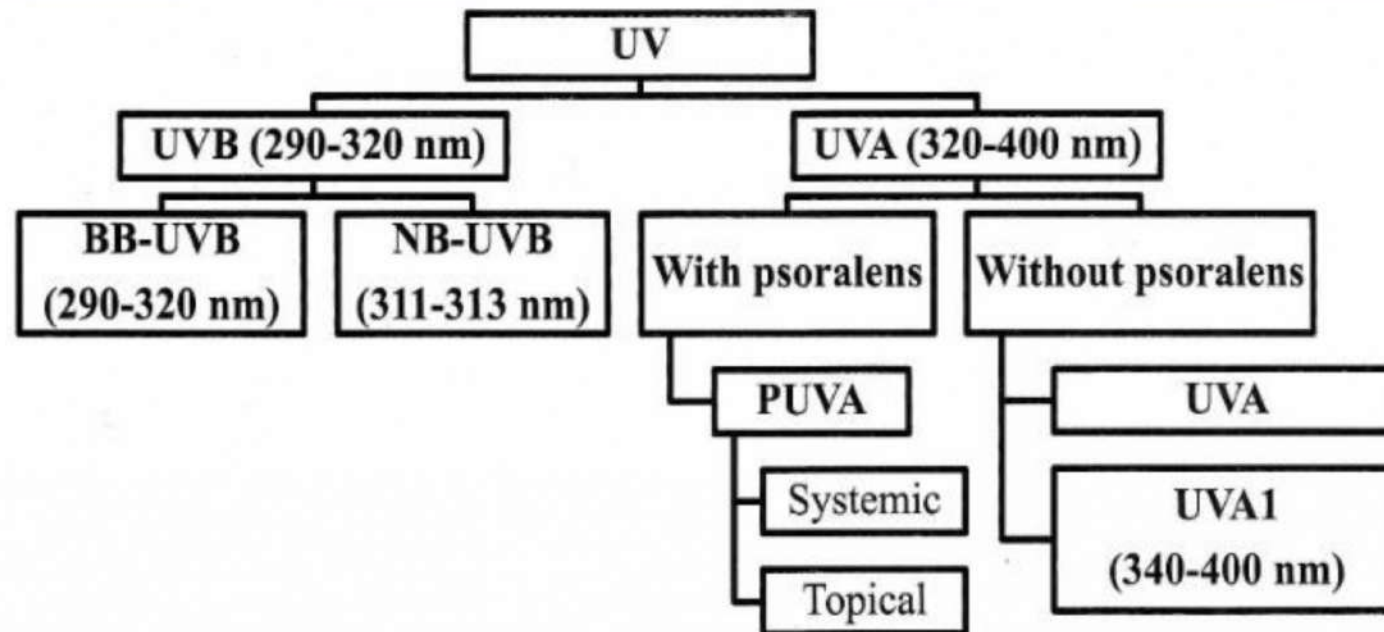
	infliximab	adalimumab	etanrecept	ustekinumab	secukinumab
	TNF	TNF	TNF	IL12 -23	IL17 A
	remicad	humera	enbrel	estelara	cosentyx
DOSE	5mg/kg , slow iv infusion	80 mg loading dose then 40mg every other wk SC	50mg twice/wk For 3 months then once weekly SC	45mg 0-month- every3months SC	300mg every week for 4 weeks then every month SC
S.E	ANAPHYLAXISI SCE Dermatology / EBDV by dr :Eman - CoNNect Academy		Injection site reaction	MF , strok	Crohns , candidiasis

C-PHOTOTHERAPY

- Definition: Phototherapy is the use of ultraviolet radiation (UVR) for the treatment of skin diseases.
- Photochemotherapy means the combination of a photosensitizer (e.g. psoralens) with UVR.
- Phototherapeutic modalities: (BB = broadband, NB = narrowband, P = psoralen)







► **Indications (in psoriasis):**

- Moderate-to-severe psoriasis not responding to topical therapies - may be used alone (monotherapy) or in combination (with topical / systemic agents).



Mechanism of action:

- Anti-proliferative: inhibits keratinocyte proliferation.
- Immunomodulatory: depletion of Langerhans cells (APCs) and shift in cytokine profile (from Th1 to Th2).

➤ **Side effects:**

- Acute: erythema, burn, tanning, reactivation of latent viral infections and photosensitivity.
- Chronic: photoaging and increased risk of cutaneous cancer.
- Side effects of psoralens: nausea, vomiting, hepatotoxicity and increased risk of cataract (PUVA).



Precautions: wearing protective goggles and genital shields during treatment.

➤ **Contraindications:**

- Photosensitive dermatoses (e.g. lupus erythematosus or porphyria)
-
- Genetic disorders characterized by increased photosensitivity or an increased risk of skin cancer (e.g. xeroderma pigmentosum)
-
- Conditions characterized by increased risk of skin cancer:
 - o Current / Previous history of skin cancer
-
- o Treatment with cyclosporine or other immunosuppressive medication

► **Baseline monitoring:**



- ANA / anti-Ro (if indicated)
- Liver function tests and eye examination (PUVA)
- Screening for cancer (if indicated)

TREATMENT OF PSORIASIS IN THE SETTING OF COMORBIDITIES

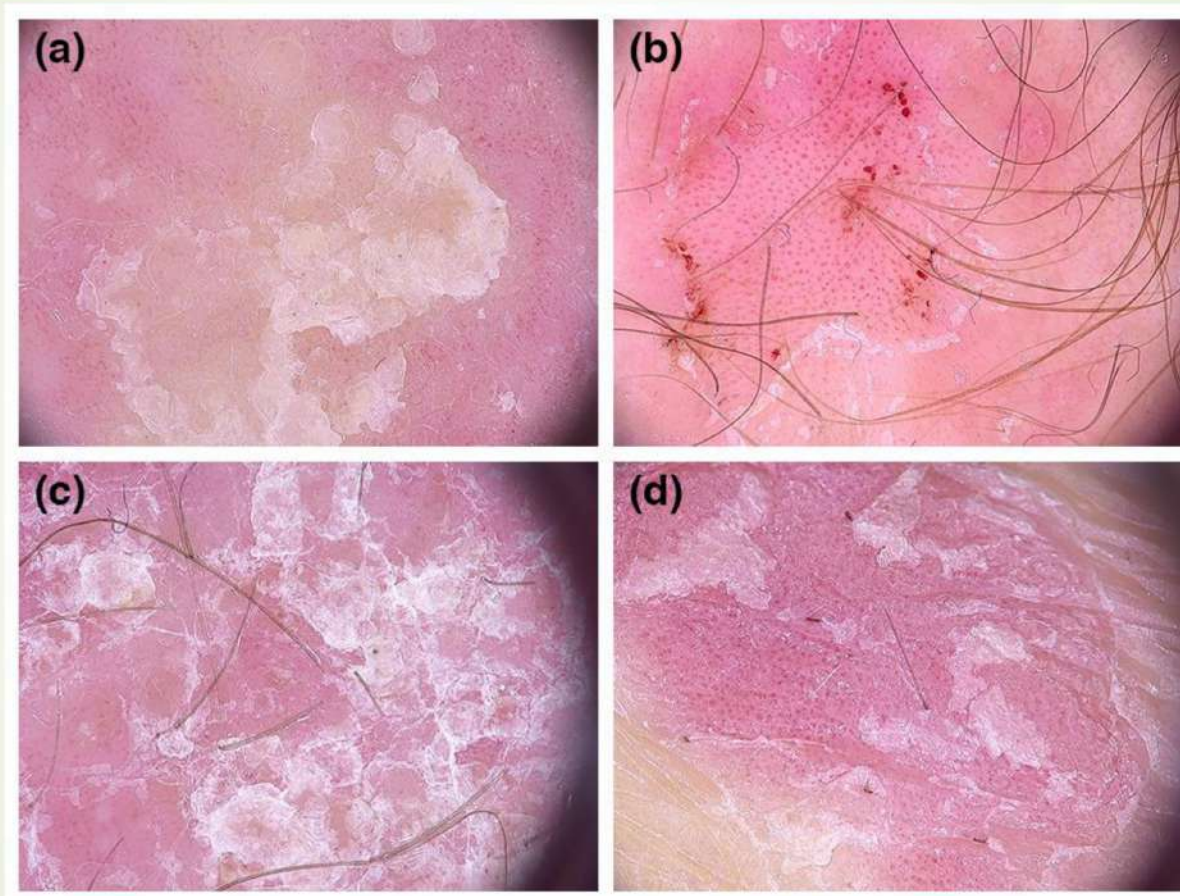
Comorbidity or special situation	Preferred treatments						
	Topical therapies		Phototherapy	Systemic therapies			
	Corticosteroids	Other topical agents		Retinoids	MTX	CSA	Immunomodulators ("biologics")
Liver disease	√	√	√ (UVB)				√
Hepatitis B* or C viral infection	√	√	√ (UVB)				√ (etanercept > adalimumab, ustekinumab > secukinumab)
Metabolic syndrome	√	√	√	√**	√**	√	√
Immunosuppression (e.g. HIV/AIDS)	√	√	√ (UVB)	√			
Pregnancy	√ (mild to moderate strength)		√ (UVB)			√ (pregnancy category C)	√ (TNF inhibitors, ustekinumab, secukinumab; pregnancy category B)
History of internal malignancy (e.g. lymphoma, lung cancer)	√	√	√ (UVB)	√			
History of frequent skin cancers	√	√ (vitamin D ₃ analogues, retinoids)		√			

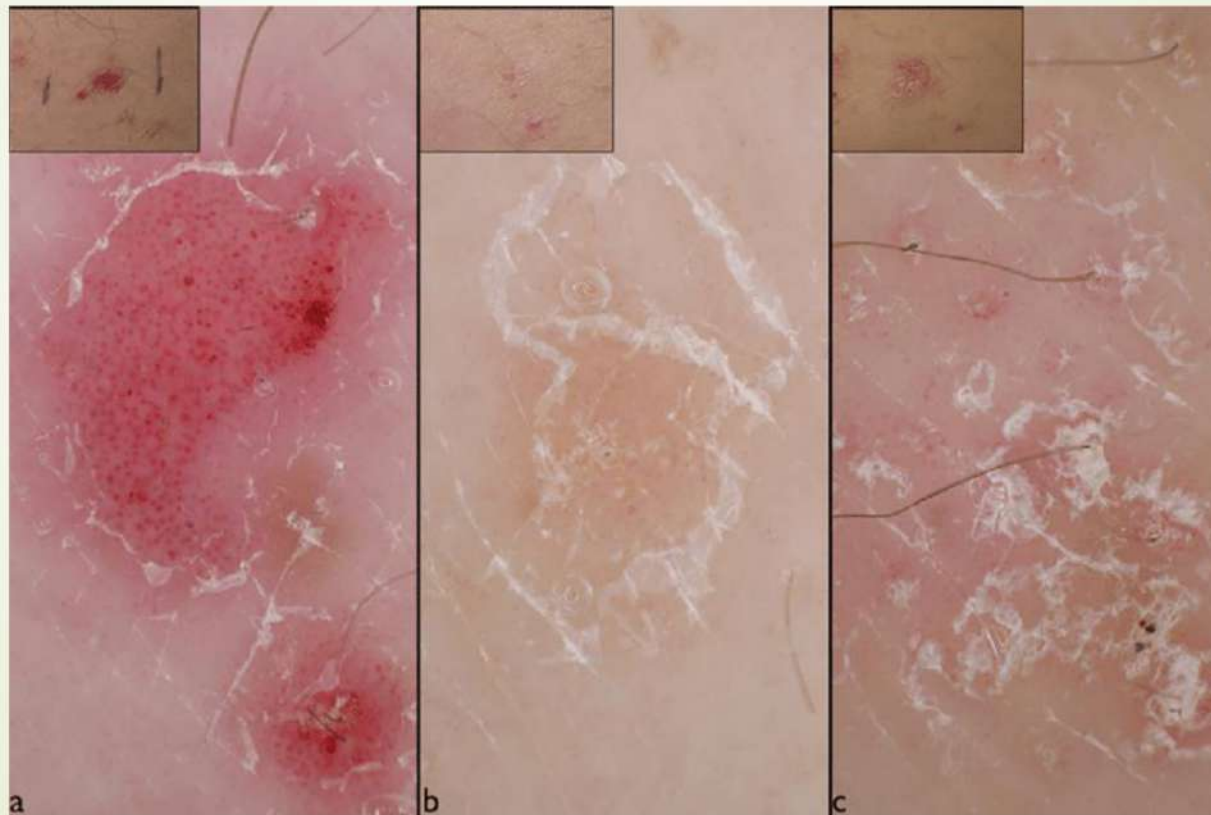
*If HBsAg-positive, measure HBV DNA and treat with antiviral agent (e.g. tenofovir, entecavir, telbivudine) prior to immunosuppressive therapy. Monitor for reactivation in those with anti-HBs or -HBc antibodies who are HBsAg-negative (see Table 128.8).

**Used with caution in patients with non-alcoholic steatohepatitis. - CoNnect Academy

Special situation	
Pustular ps	poorly-responsive to topical therapies . Systemic retinoids (acitretin) is the treatment of choice.
GPP of pregnancy	Prednison , cyclosporine
Nail ps	topical, intralesional , systemic or phototherapy) depends on (1) severity of the condition . Avoid trauma , emollient , regular trimming
Ps. arthritis	Mild : NSAID . Moderate to severe : DMARD (METHOTREXAT) Systemic corticosteroids are avoided to prevent post-steroid psoriasis flare.
Childhood ps.	Therapeutic challenge , topical as much as you can (1 st line) NB UVB , SYSTEMIC FOR SEVERE PS

What the dermoscopy of psoriasis





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